With vs. Without Document Context

Why RAG is required for reliable enterprise AI

A context-free model answers from memory; a document-grounded model answers from evidence. Using a fictional bank ("Meridian Bank"), this lab makes that difference measurable.

Baseline (no docs)

Grounded (with docs)

Enterprise takeaway

THE PROBLEM

Without documents, the model guesses

Same 3 questions about Meridian Bank, asked with no documents attached:

Q1 Max loan & interest rate?

"Personal loans typically go up to ~$50,000 at 6-25% APR..."

Risk: Generic range, no real figure

Q2 Credit-risk tiers?

"Most banks use FICO bands: 750+, 700-749, 650-699..."

Risk: Invented FICO bands - wrong system

Q3 Grace period & late fee?

"Commonly ~15 days, fees often $25-$40..."

Risk: Hedging words, no source

THE FIX — DOCUMENT GROUNDING

Same questions, now answered from the documents

WITHOUT documents

- Accuracy: generic guesses
- Source: none, unverifiable
- Specificity: vague ranges
- Hallucination risk: HIGH
- Enterprise use: unusable

WITH documents

- Accuracy: exact, matches policy
- Source: cites section (2.1, 1, 3.2)
- Specificity: TL 750,000; MRS >= 80
- Hallucination risk: LOW
- Enterprise use: audit-ready

Key difference

Grounding didn't make the model smarter - it changed WHERE the answer comes from: from statistical memory to a citable source.

WHY IT MATTERS

Why RAG is required for enterprise

- **Fluent is not the same as correct.**
  Context-free models optimize for plausible-sounding answers, not verifiable ones.
- **In finance, a wrong number is a risk.**
  A plausible-but-wrong loan rate or grace period is a compliance and financial risk.
- **RAG makes answers traceable.**
  Retrieve the authoritative document, force the model to answer from it, and cite the source.

This is exactly why my Financial Risk Analyst Agent pairs retrieval with hallucination detection - in finance, "sounds right" is not good enough; the answer must be traceable.